AI Assisted Vehicle Detection and Enumeration Using UAS

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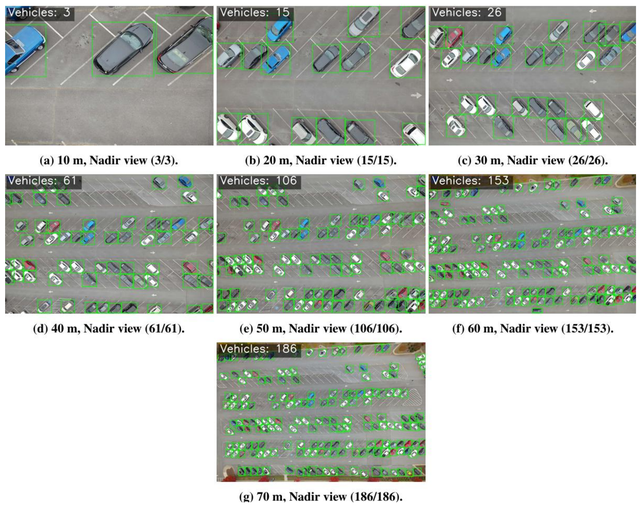
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Introduction

- Computer vision is increasingly important for transportation analytics, enabling vehicle detection, tracking, and counting using automated visual systems
- Integrating Real-time object detection with UAS to classify and localize objects offers high altitude imaging that reduces occlusion and expands monitoring capability [1]
- Aerial imaging introduces new challenges such as small object sizes, complex backgrounds, and varying camera angles [2]
- YOLO models are preferred for UAS-based detection due to their high speed, strong accuracy, and low computational cost [2]
- Traditional vision counting methods struggle with moving cameras and high-traffic conditions, challenges that are solved using YOLO

Method

- The VisDrone dataset was used to train the YOLOv12s model on images of various vehicles, using the classes car, truck, and bus.
- The fully trained model was tested on the held-out Visdrone test set
- The model was then tested on real aerial parking lot images with dense vehicle grids, mixed vehicle types and sizes, high altitude views, and challenging lighting scenarios
- The model was deployed on the SkydioX10 UAS with live video streaming to count vehicles in parking lots.
- The accuracy of the enumeration was tested on real world scenes, and the model was put under a stress test to evaluate its performance at increasing altitudes and views



(a) 10 m, Nadir view (3/3). (b) 20 m, Nadir view (15/15). (c) 30 m, Nadir view (26/26).

(d) 40 m, Nadir view (61/61). (e) 50 m, Nadir view (106/106). (f) 60 m, Nadir view (153/153).

(g) 70 m, Nadir view (186/186).

Parameter	Value
Model Variant	YOLOv12s (pretrained)
Epochs	500
Batch Size	16
Image Size	640 × 640
Optimizer	SGD (YOLO default)
Learning Rate Schedule	Cosine Decay
Dataset	VisDrone Vehicles (custom split)
Device	NVIDIA GeForce RTX 4070 Laptop GPU
Total Training Time	~48 hours

Results

- After training the model reached a precision of 0.8873 and a recall of 0.7269 on the withheld Visdrone dataset
- The accuracy of the enumeration on real world static scenes achieved a maximum error of 4% and a minimum of 0.82%
- A Nadir view consistently performed better than tilted views, with the model performing perfectly across the stress test
- Tilted views performed consistently worse, but remained below 7.5% error

Altitude (m)	View Angle	Ground Truth	Predicted	Error (%)
10	Tilted	181	176	2.76
20	Nadir	50	51	2.00
30	Nadir	52	52	0.00
40	Nadir	121	121	0.00
50	Nadir	48	48	0.00
60	Tilted	81	87	7.41
70	Tilted	293	288	1.71

Conclusion

- Developed a real-time aerial vehicle detection and enumeration system using only YOLO12vs integrated with UAS imagery
- Model achieved strong performance with high precision, high localization accuracy, and low counting error
- Real-world tests confirmed reliable real time inference and consistent detection across varied environments and vehicle densities
- Demonstrated that lightweight deep-learning models can effectively support autonomous traffic analysis and infrastructure monitoring
- Identified that models could improve on handling occlusion, extreme scale variance, and fast-moving scenarios
- With further optimization, the system can play a major role in future intelligent transportation and monitoring solutions

References

[1] Nex, F., Armenakis, C., Cramer, M., Cucci, D. A., Gerke, M., Honkavaara, E., Kukko, A., Persello, C., and Skaloud, J. "UAV in the Advent of the Twenties: Where We Stand and What Is Next." ISPRS Journal of Photogrammetry and Remote Sensing, Vol. 184, 2022, pp. 215–242. https://doi.org/10.1016/j.isprsjprs.2021.12.006.

[2] Sun, W., Dai, L., Zhang, X., Chang, P., and He, X. "RSOD: Real-Time Small Object Detection Algorithm in UAV-Based Traffic Monitoring." Applied Intelligence, 2021. https://doi.org/10.1007/s10489-021-02893-3.

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